Companion papers: **The Shape of Gravity** (the law) · **The Warp Drive** (the mechanism) · **The Hover Disk** (the materials) · **The Hexalock** (the vehicle) · **Antigravity Devices** (the bench floor) · **Superconductivity** (the coherence science)

Electrogravitics

An engineering field guide — from Townsend Brown's capacitors to a coherent-substrate drone: what is measured, what is derived, what is open, and how a drone company would actually get started.

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> **Claim boundary.** This paper makes no new physics claim. Its job is to be the single honest on-ramp to a scattered literature: it states, with the original numbers, (1) what the electrogravitics lineage actually measured — which in air is ion wind, fully explained by conventional physics; (2) what the published vacuum record bounds — every published, independently instrumented vacuum test is a null with quantitative bounds, while the one claimed vacuum positive (the 1955–57 French program) remains open, its numbers never having reached circulation; (3) what the companion papers derive — a specific force law with exactly one open number; and (4) the cheapest experimental program that would move that number, in the order that keeps a builder from fooling themselves. The coherent-coupling channel this program tests is an *unproven hypothesis, currently null on every substrate measured*. Nothing here licenses a product claim.

Every claim below carries one of six tags: MEASURED (instrument, published, or receipted in this project) · DERIVED (follows from the framework's law, receipt named) · ESTIMATE (a prior, honestly labelled) · PRE-REG (frozen prediction, not yet run) · OPEN (the bench decides) · NULL / REFUTED (tested and failed).

PART I — THE INTUITION

1. The word, and what was actually measured

Electrogravitics (that is the standard spelling) is Thomas Townsend Brown's word, from the 1920s, for the claim that a strongly asymmetric, high-voltage capacitor feels a net force toward its small electrode — and that this force is gravitational in character, not electrical. The lineage is real and documented: Brown's patents (GB 300,311 of 1928 through US 3,022,430 of 1962), the Project Winterhaven proposal to the military (1952), and the *Electrogravitics Systems* survey (Aviation Studies International, February 1956) that briefly made "electrogravitics" an aerospace buzzword. MEASURED

Here is what six decades of instrumentation established about the force Brown measured:

- **In air, the force is ion wind** — electrohydrodynamic (EHD) thrust. Charge is sprayed off the sharp electrode, drifts to the blunt one, and drags neutral air with it. The thrust obeys `T ≈ I·d/μ` (current × gap / ion mobility, $\mu \approx 2\times10^{-4}$ $m^2V^{-1}s^{-1}$) — it scales with *current and gap*, not voltage, which is the fingerprint. Measured lifter thrusts reproduce this to within geometry factors of 0.1–0.8 (Bahder & Fazi, ARL-TR-3005, 2003 — often miscited as supporting electrogravitics; it is an ion-wind paper). MEASURED
- **Ion wind is genuinely useful** — MIT flew a solid-state airplane on it (Xu et al., *Nature* 563, 2018: 2.5 kg, 5 m span, 40 kV, ~6.25 N/kW). It is also intrinsically atmospheric: *zero thrust in vacuum*, and human-scale EHD lift needs ~330 m^2 of electrode per person. A real technology, the wrong tree. MEASURED
- **In vacuum, every published, independently instrumented test is a null.** Talley (AF-funded, 1988–1991, 10^{-6} torr): no thrust except during breakdown current. Tajmar (2004): the residual force was isolated corona wind. Bartlett & Tajmar (*Acta Astronautica* 181, 2021): ±0.3 mg resolution to 10 kV, no force in any configuration. The one claimed exception — the 1955–57 SNCASO "Project Montgolfier" vacuum residual — is a corporate-documented positive whose numbers have not survived into circulation; see its §3 ledger row. NULL (published record)

> **The verdict, stated first because everything else depends on stating it first:** Biefeld–Brown electrogravitics, as an antigravity claim, is *refuted*. The force Brown measured in air is ion wind. This paper's own framework does not fight that verdict — it *requires* it (§4): an incoherent capacitor is predicted to couple to spacetime at the bare Einstein-constant floor, $\sim10^{-43}$, unmeasurably below every balance ever built. Any program that starts by disputing the nulls is not doing engineering. This verdict is scoped to what was measured: the atmospheric force — and the Montgolfier project's own 1959 report agrees, calling it "a pure demonstration of 'electric wind.'" The French *vacuum* residual is a separate, open question (§3

> ledger): a documented claimed positive whose numbers have not survived into circulation, testimony on its behalf included. Testimony raises retrieval priority; only the numbers — or the §6 bench — can move the tag.

So why is the word worth keeping? Because Brown got one thing right that the debunking, correctly aimed at his interpretation, never touched: **the geometry**. An asymmetric capacitor is a device for storing field energy with a built-in spatial gradient — a shape that stores energy on one side of itself. In the framework of the companion papers, that is precisely the shape a gravitational coupling would need (§2). Brown had the right blueprint and the wrong active ingredient: high voltage on an *incoherent* dielectric, where the framework says the missing ingredient is *coherence*. That diagnosis is falsifiable and cheap to test, which is what Part III is for. OPEN

2. Sixty seconds of gravity, and what "engineering it" would mean

Start with the part that is not a hypothesis: **electromagnetic fields curve spacetime.** That is Einstein–Maxwell — coupling $8\pi G/c^4$, no free parameters — and it has been solved *exactly* for laboratory hardware: Füzfa (*Phys. Rev. D* 93, 024014, 2016) computes the full nonlinear spacetime curvature of real current-carrying coils (effect $\propto I^2$, i.e. $\propto B^2$), with a proposed interferometric detection. A field-conditioned machine *does* modify local geometry, unconditionally. THEOREM The honest arithmetic that goes with the theorem: Füzfa's CERN-magnet-class stack shifts an optical phase by 1.6×10^{-25} rad per round trip (10^{-11} rad only after 200 days of coherent integration), and at 100 T the Melvin curvature radius is 3.7 light-years. The mechanism is settled; the entire engineering question is magnitude — the ~40-order climb from the bare coupling to a useful one.

The mechanism story is owned by The Shape of Gravity; here is the engineer's summary, with the receipts named. In this framework spacetime is a network of elastic edges, and a mass is stable energy settled into it. One derived law with no free parameter — `F = 1 + E/E★`: the effective length of an edge grows linearly with the energy it carries — turns out to close all of weak-field gravity. Settled energy lengthens nearby edges (space curves) and slows clocks on them (time dilates), the same δ doing both, which forces Einstein's full factor-two light deflection with no tensor added: the ray-trace converges to 2.000 (`crypto_oracle/shape_F_gravity_closure_probe.py`), and the 1/r Newtonian halo fits at $R^2 = 0.9998$. DERIVED What is *not* derived is the absolute strength — Newton's G — and the paper says so on purpose. OPEN

> **The claim has three layers, and only one of them is a bet.** (1) Mechanism — theorem-level, unconditional: fields gravitate. (2) Bare-coupling magnitude — settled, and bracketed from above by the null program (§3). (3) Coherent enhancement — the open bet the §6 bench decides. The correct register is therefore "**this works; the climb is the program**" — exactly as fusion's question is confinement, not whether fusion works. The closest respectable published cousin of layer 3 is Ummarino & Gallerati's superconductor gravito-Maxwell work (EPJ C 2017 → Front. Phys. 2022: transient $\Delta g/g \sim 10^{-7}$–10^{-5} near T_c — screening, not generation, untested), which shows a peer-reviewed literature reaching for the same lever at far more modest magnitude.

If gravity is "settled energy sets effective geometry," then engineering gravity means one thing: **store a lot of coherent energy in a body, asymmetrically, so that the body sits in a geometry gradient of its own making — and couple to the lattice well enough that the gradient does mechanical work.** Three consequences follow immediately, and they organize everything a builder needs to know:

1. **Hover is statics; travel is dynamics.** A static force that balances weight does no work ($P = F \cdot v = 0$). If the coupling exists at all, holding position costs only losses — like a table holding a book, or a maglev rail holding a train. Accelerating costs real energy, and a conservative coupling makes deceleration regenerative. The lawful summary is "hover ≈ free, momentum costs work" — *not* "free energy." DERIVED
2. **Propellantless is not reactionless.** Every force needs a reaction partner. A propeller throws air; a rocket throws mass; this coupling, if real, hands momentum to the lattice — the same bookkeeping as maglev, where the rail takes the reaction. "Where does the reaction momentum go?" is the first question to ask of any device claim, ours included. A body that rigidly carries its own static field configuration is a closed loop and gets exactly zero net thrust — confirmed to 10^{-17} N in simulation under both classical force laws (`crypto_oracle/universe_sim/ampere_tension_sim.py`). You cannot fall into a pocket you carry. DERIVED
3. **Symmetry gives zero.** A symmetric energy store makes a symmetric dent and no net force. All thrust lives in the asymmetry — in the drive's spatial gradient (the

Brown geometry), or its chirality (swirl/spin), or its time-asymmetry (transient "jerk" drive). This is why the sign-reversal test (§6) is the cleanest discriminator: flip the asymmetry, and a real coupling flips sign; every artifact doesn't. DERIVED

> "**Lock, not lift**" **— the phrase a drone company should internalize.** A propeller *lifts*: it pumps momentum downward continuously, and when it stops you fall. A working coherent-coupling disk would *lock*: engage the coupling and the lattice pins the pose, at near-zero power, the way a rail pins a maglev. Thrust is then a controlled *modulation* of the lock — continuous through zero, sign set electronically in microseconds, no reversal transient, regeneratively recoverable. Every one of those properties is exactly what 6-DOF multirotor control wants and cannot have with propellers (§5).

How to picture it — five images that hold, and three that don't

A century of this literature has produced a handful of genuinely good mental pictures and a handful of seductive wrong ones. Both are worth having, because the wrong ones are the ones your team will meet first.

The images that hold. *Gravity does not pull; it adjusts the characteristics of space* — the best one-line statement of general relativity in the whole corpus, and it comes from the proponents' own 1956 *Electrogravitics Systems* report. *There is no perfectly rigid rod*: near a mass, rulers shrink and clocks slow, so a circle's circumference comes out less than 2πr — that is what "curved space" operationally means (Dicke's and Puthoff's medium picture, and it is mainstream). *Light bends near a mass the way it bends entering glass* — spacetime as an optical medium of varying index, a picture as old as Eddington. *A moving mass makes a gravity analogue of a magnetic field* — the spinning Earth drags spacetime like a spoon through honey, measured by Gravity Probe B at ~39 milliarcseconds per year, which honestly sets the scale of what mass currents do. And the one that governs the ledger: *the ion lifter is an invisible propeller made of air* — the wire is the blade you cannot see, each ion is a runner in a bucket brigade handing momentum to neutral air, which is exactly why the thrust scales with current, ignores which way the voltage is wired (a fan doesn't care which way the motor is wired), and dies dead in vacuum: no one to hand off to.

The images that mislead — teach them, then break them. "*Matter is just an electrical condition, therefore electricity is connected to gravity*" (Brown, 1929, the ancestor of every argument since): the premise is true — mass-energy includes field energy, which is precisely why a charged capacitor weighs imperceptibly more — but "connected at 10^{-43}" is not "controllable," and the sentence smuggles one into the other. *The surfboard on a self-made hill* (Mason Rose, 1952; the single most memorable image in the field): a real surfer rides a slope sourced by the Earth and maintained by the ocean; a craft that makes its own hill and carries it along is pushing its car from the inside — the reaction has to land somewhere, and if that somewhere is the air you have built an ion thruster. Our own conservation books say the same thing in equations: a symmetric, carried pattern gives exactly zero (§4). "*Ion wind is three orders of magnitude too small*" — a half-quotation of the Army Research Lab paper whose *other* half computes ion drift at the right magnitude, after which NASA (2004) and Ianconescu (2011) turned it into a full model that predicts the thrust, its ground-wire dependence, and its polarity independence. And the one image to keep from the proponents because it is almost right for the wrong reason: the 1956 report's *electrostatic wing* — "the same lift as pressure and suction over a wing, except that no airflow is necessary." Right shape; the one wrong clause is the entire lesson.

3. The measured ledger

One table, so nobody has to reconstruct the evidentiary state from folklore. "Bound" is the constraint the measurement places on the framework's one open coupling number (§4).

Item	Record	Status
Ion wind (EHD) thrust in air	$T \approx I \cdot d/\mu$; lifters, MIT 2018 airplane; ~0.15 mN/mA at 3 cm gap	MEASURED — real, atmospheric-only, conventional
Biefeld–Brown force as antigravity	Explained by EHD in air; vacuum nulls below	REFUTED
Talley 1988–91 (USAF), 10^{-6} torr	No thrust except during breakdown current	NULL
Tajmar 2004, 38 kV, corona-excluded	$<10\ \mu N \rightarrow$ coupling $< \sim 2\times10^{-3}$	NULL

Bartlett & Tajmar 2021, ≤10 kV	nano-N resolution, no force → coupling < ~2×10^{-10} — **the tightest published bound**	NULL
1955–57 Paris tests ("Project Montgolfier", SNCASO, facilitated by J. Cornillon; final report April 15, 1959)	Documented corporate program, archive released by Cornillon before his death (2008; Starburst/LaViolette overview 2012; primary site now offline). Its own report attributes the *air* motion to "a pure demonstration of 'electric wind'" — agreeing with this ledger. The vacuum tests (1.4 m chamber, 5×10^{-5} mm Hg) reported a small *polarity-locked* residual ("discs always moved toward the positive pole") called "a definable force" — but no thrust value, voltage–force curve, or current log circulates, and a polarity-locked residual amid sparking has a named conventional candidate (charge-emission / vacuum-arc momentum: Martins & Pinheiro 2011; cf. Talley's thrust-only-during-breakdown). Decisive test the record lacks: simultaneous current + force	OPEN THREAD — claimed positive, documents released but numbers not yet retrieved
Buhler / Exodus Propulsion (asymmetric dielectric capacitor stacks)	Patent's one measured number (237 μN) was in air — inseparable from ion wind; the mN-scale vacuum figures are talk-claimed (~2000 runs), unreviewed, unreplicated — and declining: the claimed force fell ~40× between Apr 2024 (1 g on a 30–40 g device) and Mar 2026 ("5–10 mN," rebranded "electrostatic pressure force") with no published explanation — the monotonic decline-under-scrutiny signature — still with no paper and no independent instrumented test, while the claim sits 10^5–10^6× above the Dresden noise floor in the same HV regime. Back-solving our force law on the claimed numbers implies a coupling ≈ 2×10^{-4} — consistent within an order of magnitude, *not validated*, and ~125× below the hover threshold	OPEN THREAD — watch, don't bank; declining
Tajmar, Kößling & Neunzig 2024 (*Sci. Rep.* 14:19427): broad gravity–EM coupling sweep	Capacitor geometries, solenoids, tunneling currents, varistors, ε/μ gradients, crossed & helical B, shielded toroids — all bounded at **~nN force / nN·m torque** in high vacuum; supercurrent→gravity bounded at 5×10^{-9} N/A (*Front. Phys.* 2022). The tightest map yet of which shortcuts do not exist	NULL (comprehensive)
Podkletnov rotating-	NASA Marshall and Göde replications null to 0.001%	UNREPLICATED

superconductor weight loss		
Li–Torr gravitomagnetic amplification (theory)	Refuted on its own terms (Harris 1999); AC Gravity LLC published nothing	REFUTED
Graham et al. 2008 (Canterbury), rotating s-wave SC ring laser	The cleanest existing upper bound on lattice coupling in ordinary superconductors	MEASURED bound
Pais "HEEMFG" (Navy, 2016–19, ~$500k)	"Could not be proven" — but the rig could spin *or* vibrate, never both, and never reached the specified charge: an uninterpretable null, not a refutation. Mine the patents for geometry only (§7)	NULL (weak)
Wallace patents US 3,626,605/606 (GE era, 1971): rotating half-integer-nuclear-spin masses	13-arcmin torsion claim; unreplicated — and, unlike Brown, **no published null either**	OPEN THREAD (§8)
This project's own bench	No substrate measured to date has shown the coupling; every xv-channel measurement so far is null	NULL to date

Read the last row again before reading Part II. The framework's force law is a *prediction under test*, made by people who publish their own nulls. That is its credential.

PART II — THE ENGINEERING FRAME

4. The force law, for engineers

The mechanism paper (The Warp Drive) and its math audit (`WARP_DRIVE_MATH_AUDIT.md`) give the working force law. As pressure on a driven substrate of area A:

```
P_w  =  C·κ·½ ε_r ε₀ E²        [Pa]         (field / capacitor drive)
F    =  C·κ·U_drive / L_grad    [N]          (general stored-energy form)

C  ≡  C_sub·|χν|²  — the lattice coupling           ← THE open number
κ  ∈ [−1,1]        — drive asymmetry (0 = symmetric = zero force; sign(κ) = sign(thrus
U_drive            — coherently stored energy (not dissipated power)
L_grad             — gradient length ≈ device radius
```

> **What C actually is — the shout and the room.** Think of the drive's stored energy as a *shout*, and the lattice — spacetime's substrate — as a *room that can echo*. How much force comes back depends on two independent things, and that is why C is a product of two factors. **$|\chi\nu|^2$ is the tuning**: how well the substrate's own coherent mode overlaps the lattice's mode — matched frequency, matched geometry, phase locked across the whole body. It is dimensionless, sits between zero and a hard geometric ceiling (the derived vertex-sharing bound, $g \le 1/\sqrt{3} \approx 0.577$), and enters squared because coupling is an amplitude and force follows energy — the same reason a wave's power goes as amplitude squared. A bronze cymbal, driven at audio, tunes at $|\chi\nu|^2 \sim 10^{-14}$; an optically driven diamond disk is estimated near 10^{-1}. **C_sub is the room's echo efficiency**: given that the modes overlap at all, what fraction of that overlapping energy the *substrate* converts into force — how much of the body participates coherently, how much energy it can hold before it cracks or breaks down, how consonant its structure is with the lattice. It is the honest name for everything about the *material* that is not the mode overlap. They are kept separate because they fail independently and are fixed independently: perfect tuning on a body that cannot hold energy is nothing, and a superb energy store driven at the wrong frequency is nothing — force needs both, so the materials program is two ladders that multiply. One number is what a balance reads out; splitting it into two is the framework's claim about what that number is made of, not a second measurement. And the scale to hold in mind: an incoherent lump — Brown's capacitor — sits at the bare-vacuum floor of C, the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$, while hover needs $C \approx 10^{-6}$. That forty-order climb is the whole job, and both factors are levers.

Hover is the pressure condition `P_w ≥ ρ·t·g` (areal weight) — and because area cancels, **self-lift is size-independent**: a property of material and drive, not diameter. Scale anchors, all DERIVED from the law plus known material constants:

- A 1 mm diamond-class skin weighs ~**34 Pa**. Full electrostatic pressure at 1 GV/m in diamond is ~25 MPa — so hover needs the coupling·asymmetry product to deliver only ~10^{-6} of the stored pressure. On the coherence dial of the audit:

hover at S_coh·f_cons ≈ 1.4×10^{-6} — about one part per million of full coherence.

- The consonance ceiling is f_cons ≤ e^(−P/12) ≈ 0.873; the bare-vacuum floor — what an *incoherent* body couples at — is the Einstein constant $8\pi G/c^4 \approx 2\times10^{-43}$. The whole game is how far real coherence climbs a body up those ~40 orders of magnitude. OPEN
- The tightest published null (Bartlett & Tajmar 2021) bounds *statically driven* capacitors at $C < 2\times10^{-10}$ — which is 33 orders above the floor and 4 orders below Buhler's implied value. The framework's own explanation for why the nulls don't close the question: every null device to date was a static, incoherent capacitor, which by the framework's own predicates cannot access the coherent channel. That is self-consistent, and it is also the framework rescuing itself with its own predicate — the papers say so plainly, and the §6 program is designed to close exactly that loophole rather than lean on it. OPEN

4.1 The levers

Lever	What it is	Status
Stored energy U_drive	Coherently stored, not dissipated. Material ceiling is specific resilience $R = \sigma^2/(E\rho)$ for mechanical storage, $\frac{1}{2}\varepsilon_r\varepsilon_0 E^2$_bd for field storage. Diamond is the single-material champion on both counts; graphene's ceiling is ~7000× higher again but its coupling velocity is genuinely uncertain	DERIVED
Asymmetry κ	The Brown geometry: small/large electrode, domed top over larger bottom shell, insulation ring — an asymmetric capacitor is κ rendered in hardware. The asymmetry "wants to be a swirl": chiral electrodes, spin, and time-asymmetric (jerk) drive are the same lever in space, rotation, and time	DERIVED (form) / OPEN (magnitude)
Coherence S_coh	The claimed missing ingredient. Fraction of the body participating in one phase-locked mode: Q, isotopic purity (^{12}C, ^{28}Si), condensate boundaries, cold. See Superconductivity for the measured science of macroscopic phase coherence	OPEN

Consonance f_{cons}	Geometric match between the substrate's mode pattern and the lattice (icosahedral symmetry, φ-ratios, degree-3 carbon cages, quasicrystals). Ceiling 0.873	OPEN
Frequency	**Deliberately listed last, because it is the fork, not a lever.** Hypothesis A: coupling rises with drive frequency ($\eta \approx v_{phase}/c$ — THz/optical drive wins by 8–12 orders). Hypothesis B: coupling is stored-field-limited and DC suffices (Buhler's DC claims point here). A frequency sweep on one device decides it (§6). Do not design a program that assumes either answer	OPEN — the A/B fork

What a null bounds, and what it doesn't. A vacuum null on your device bounds $C < \delta F \cdot L_{grad}/(\kappa \cdot U_{drive})$ for *your* drive condition. It refutes the mechanism only if the chamber-variation and frequency-sweep controls also come back flat. And know exactly what the published null program covers: it — including the comprehensive 2024 Dresden sweep — is excellent metrology, and it is binding *for what it tested*: static fields, DC currents, incoherent substrates. No published test article has carried even one of the four §4.1 coherence predicates (lattice-matched symmetry, coupling channel, condensate boundary, near-mode drive), and a static — or uniformly rotating, mirror-symmetric — drive has $\kappa = 0$ by the conservation books: it structurally cannot rectify. This is worth stating as a two-framework agreement: for every article the null program tested, general relativity predicts effectively zero (weak-field arithmetic) *and this framework predicts exactly zero* (symmetric drive, $\kappa = 0$, closed loop) — the same weak-field regime our companion law re-derives. The two theories part company only where the coherence predicates are met and the drive is rectified; nowhere tested so far is in that regime. So the test matrix has exactly one untested cell, and the coherent-substrate architecture of the companion papers sits in it. That is not a rescue; it is a target. The §6 bench is the first experiment that would fill the cell, and a null *there* — with the coherence predicates instrumented and verified present — would be the first null that actually reaches this framework. Conversely — and this is the discipline that keeps you honest in the other direction — a *positive* force on your balance is an artifact until it survives: vacuum, thermal null, dummy-load null, sign reversal under drive flip, chamber variation, and sidereal folding. The published field is a graveyard of people who skipped one of those.

5. Why a drone company is the right builder

It is tempting to think this technology, if real, belongs to aerospace primes. The opposite is true, and the reasons are concrete:

1. **The control problem is already solved, and it's yours.** A craft with six independently throttleable force patches is a 6-DOF wrench-map inversion — exactly the mixer a multirotor autopilot runs today. The Hexalock paper works this out on the published Lynchpin airframe (Howard et al., SAE Int. J. Aerosp. 16(3), 2023 — a 651 g prototype that *flew*, on PX4, with six bidirectional props): six disks on the six pentagons of a collapsed dodecahedron, a 6×6 always-invertible mixer, one RP2350 whose 12 PIO state machines map one-to-one onto 6 disks × {amplitude, κ-steer}. Swapping the thruster type does not touch the mixer. The flight-control ledger is closed; only the lift ledger is open — and a drone company is precisely the builder for whom the closed half is free. MEASURED (the airframe/control half)
2. **The disk fixes your hardest actuator problem.** 6-DOF vehicles need signed, arbitrarily small thrust per axis — the constraint that pushed ETH's Omnicopter to eight over-actuated rotors and the Lynchpin prototype to bidirectional props with reversal latency. A coupling disk's thrust is continuous through zero with electronically-set sign: no mechanical state, no reversal transient. DERIVED (from the law's κ-linearity — conditional on the coupling existing at all)
3. **Endurance inverts.** A propeller drone sizes its battery for hover; a lock-mode craft sizes it for *maneuvering*, because station-keeping costs only losses. The first real product isn't a faster drone — it's a drone that holds station indefinitely on a trickle. DERIVED (conditional, same caveat)
4. **The bench is drone-shop-sized.** Nothing in Part III needs a wind tunnel or a cleanroom: a vacuum bell jar, a torsion or pendulum balance, a HV supply, an RP2350, and discipline. The single most dangerous item is the 40 kV supply, which drone shops handling LiPo banks already respect.

> The strategic shape: everything above the coupling is flight-proven engineering you already do; everything below it is one number. Build the program that measures the number, and structure it so that every stage yields a sellable conventional capability (ion-wind actuation, HV competence, precision force metrology, THz materials) even if the number stays zero.

6. Getting started: the bench program

Rule 0: bench one disk before you build six. The Lynchpin paper states it as a failure mode: the headline risk is that the disk does not lift at all, so never invert the order. Everything below is one disk on one balance.

Stage 0 — Calibrate on physics that works (weeks, ~$1k)

- **Build the ion lifter first.** Fly it, measure T vs I·d/μ, watch it die in the bell jar. Now your team owns the artifact that contaminates this entire field, and your instruments are proven on a real force. MEASURED
- **Diamagnetic control:** pyrolytic graphite levitating over NdFeB — zero-power levitation on textbook physics ($B \cdot dB/dz = \mu_0 \rho g/|\chi|$). This pins the conventional baseline any later in-field force claim must be distinguished from. MEASURED
- **Balance floor:** a 2 m pendulum or torsion balance with interferometric readout reaches ~nN–500 pN. Characterize thermal drift, EMI, and electrostatic pickup *before* any candidate device goes on it.

Stage 1 — The discriminator set (the actual experiment)

One asymmetric capacitor device, in vacuum, on the calibrated balance, run through five controls. This set — not any single reading — is the experiment:

1. **Vacuum arm, always.** No in-air number means anything in this field. (The audited-and-rejected proposals share one tell: no vacuum control anywhere in the plan.)
2. **Sign reversal under drive flip.** Flip the asymmetry (polarity / phase / chirality): a lattice coupling reverses sign; heating, outgassing, and electrostatic pickup do not. The cleanest single discriminator the framework offers.
3. **κ → 0 null.** Symmetric drive at equal power must give zero. If it doesn't, you're measuring an artifact.
4. **Frequency sweep, DC → RF (→ THz where you can).** This is the A/B fork of §4.1 decided on your own hardware: force rises with frequency ⇒ hypothesis A;

flat ⇒ hypothesis B (or nothing).

5. **Chamber variation + sidereal folding.** A lattice coupling is chamber-independent; trapped charge tracks the walls. And fold every long log at both 24 h 00 m and 23 h 56 m: any real preferred-frame coupling modulates at the sidereal day, where no mundane artifact lives. Costs zero hardware on a timestamping MCU.

Any force that survives all five is worth a phone call. A null across the sweep is also a result: it is the tightest bound ever placed on the coherent channel, publishable as such.

OPEN — this is the bench that decides

Stage 2 — The first article worth building (months)

The corpus's convergent first build ("Architecture C") is deliberately shop-buildable: two facing domed copper discs, 80–120 mm, across a corrugated gap; a **chiral-spiral electrode** charged to a few kV (the spiral *is* the κ direction — no steering electronics); the shell **spun** by one balanced BLDC at governed RPM (a flywheel, not a propeller — spin supplies the high-frequency mode mechanically, which no published null device had); per-disk flyback or Cockcroft–Walton HV; passive cooling, because colder is strictly better for coherence. Rationale: every published null was a *static* capacitor; this device is the cheapest object that actually exercises the claimed channel (stored field + asymmetry + rotation + time-structure) with independent control of each.

OPEN — a build hypothesis, not a result

Stage 3 — The coherence ladder (the materials program)

Owned by The Hover Disk; the drone-relevant summary: substrate ladder steel → bismuth → $BiFeO_3$ → quasicrystal; the single-material optimum is a ^{12}C-enriched single-crystal diamond disk driven optically (structural champion *and* η ≈ 0.41 optical substrate); the layered endgame adds an icosahedral-quasicrystal symmetry layer, a THz-coupling multiferroic film, and a superconducting boundary (YBCO at 77 K) as the phase-coherence layer. Two free pre-screens: broadband absorptance (good couplers should be anomalously black — necessary, not sufficient) and the diamagnetic control from Stage 0.

> **Superconductor discipline.** The condensate boundary enters as a *coherence layer*, not as an antigravity ingredient. [The measured record]() is unambiguous: flux expulsion is a property of the superconducting state, not a road into it, and every rotating-SC weight-loss claim is unreplicated (Podkletnov) or refuted (Li–Torr). The honest statement is Canterbury 2008's: ordinary s-wave superconductors bound the coupling from above. If a condensate boundary matters here, it is because phase coherence is the claimed missing ingredient — and that is exactly what Stage 1 measures.

The desk work that grounds it

The Biefeld–Brown retrodiction sim — predictions frozen in a pre-registration before the run (`papers/BIEFELD_BROWN_SIM_PREREG_2026-07-19.md`), then executed (`crypto_oracle/universe_sim/biefeld_brown_sim.py`, receipt `runtime_receipts/biefeld_brown_run.json`, results `papers/RESULTS_BIEFELD_BROWN_2026-08-13.md`) — **passed all four frozen predictions and all three gates.** An incoherent Brown-class capacitor at the bare-vacuum floor produces at most 3×10^{-42} N in vacuum — 35–39 orders below the Talley/Tajmar/Bartlett balances, retrodicting every published null; the in-air force is reproduced by the ion-wind side-calc to within 12% with zero metric contribution; the coherent arm crosses the hover threshold at $C \approx 1.6\times10^{-6}$ (the audit's hand-derived dial value was 1.4×10^{-6}, frozen before the run); and the conservation controls (symmetric drive, carried pattern, static alignment) are zero to machine precision — the bootstrap is structurally impossible in the model, not merely penalized. **MEASURED (in-model)** Per the pre-reg's own register rules: this validates that the framework's story is *coherent and consistent with the published nulls* — it is not bench evidence that the coherent force exists. The bench (§6, stages 0–2) still decides reality.

7. Dead ends, priced — do not spend money here

- **Unruh / zero-point-field propulsion.** The one engineering proposal in the gravity paper's audit whose premise is true (the Unruh bath is textbook physics) was priced with every assumption in its favour: $T = 4.06\times10^{-21}$ K per m/s² — so no vibrating structure of any kind is a route (a 40 kHz × 10 μm ultrasonic drive makes 2.6×10^{-15} K of bath); break-even sits at 160.7× the Schwinger field, i.e. *the vacuum tears before the bath pays for itself*; and the technology-proof killer is the identity τ = 2πc/a — the bath's coherence time is set by the same acceleration that makes it, so there is no configuration in which you aren't paying for the reservoir you're harvesting. **PRICED & CLOSED** (`runtime_receipts/unruh_zpf_anisotropy_pricing.json`)

- **Free-air acoustic hover.** The 1-atm ceiling and the momentum bookkeeping exclude human-scale free-air acoustic levitation — a pressure cushion's reaction goes into the ground, so it is ground-effect only. (This project retracted its own hoverbike lift story on exactly this audit; the corrected diagnosis is in `hover/GROUND_EFFECT_DIAGNOSIS.md`.) **EXCLUDED**
- **The over-unity package deal — stated precisely.** What conservation rules out flat is *closed-system* over-unity: cyclic net work from a single equilibrium bath, no reservoir named. Devices offering lift *plus* that are the single most reliable fraud marker in this literature — refuse the package. **FORBIDDEN** What this does *not* cover: open-system vacuum/near-field energy harvesting, which is a separate lane in this project with a measured, peer-reviewed anchor (Moddel-class MIM optical-cavity devices, ~70 W/m², output ∝ 1/d) and a lawful frame — rectifying the ambient thermal near-field against the effectively-0 K zero-point sector is a Carnot pair, and its non-negotiable signature is that a genuine device **runs cold under load** while an artifact warms. See `zero_point_energy.html` §7 and `optical_cavity_zpe.html`. The discipline: never argue conservation by reflex; *demand the reservoir inventory and the thermal sign*. A device claiming lift + energy that names its reservoir and shows the cold signature has earned a bench date, not a dismissal — one that names neither gets the fraud-marker treatment. **OPEN (energy lane's question, not this paper's)**
- **Pais patents as mechanism.** The Navy patents are real and granted; the claimed physics fails by ~10 orders on its own numbers, the HEEMFG program's null stands (weakly), and the RT-superconductivity patent fails a kinematic bound a quartz watch refutes. Mine the *geometry* (double-shell charged/insulated cavity, counter-spin, acousto-optic dual drive) and discard the claims. **REFUTED as physics** / **OPEN as geometry**
- "**The strong force is amplified gravity**" (Nuclear Gravitation Field Theory and kin): fails universality — the dineutron, which its own logic predicts binds tighter than the deuteron, is not bound at all. **REFUTED**

8. The under-exploited threads

Three places where the next real datum is cheapest:

1. **The Wallace rerun.** Henry Wm. Wallace (GE, patents US 3,626,605/606, 1971) built rotating-mass devices whose materials were chosen by one axis nobody else has used: *half-integer nuclear spin* (Cu-63/65, Zn-67, Pb-207; spin-zero Fe and ^{12}C as "insulators"). Unreplicated — and with no published null either. His Barnett-effect nuclear polarization at 28 kRPM was $\sim 10^{-10}$; modern hyperpolarization (DNP/SEOP) reaches 0.1–1 — **a rerun today is nine to ten orders of magnitude stronger on the source side**, at university-lab cost. Bismuth-209 (spin 9/2, the highest of any quasi-stable nucleus) is the extremum of Wallace's own design axis and independently the corpus's rung-2 substrate. This is the best unplayed card in the electrogravitics deck. OPEN
2. **The hairpin bench.** The two classical force laws for current elements (Grassmann/Lorentz vs Ampère) agree exactly on every closed circuit and differ by a finite, cutoff-free **29%** on an open hairpin bridge closing through rails (5.35 vs 6.91 gram-force at 270 A — `crypto_oracle/universe_sim/ampere_tension_sim.py`). A supercap bank, a load cell, and a weekend settle a 200-year-old question about longitudinal forces — and the open-circuit/closed-circuit distinction is the same one that governs whether a craft can push on the lattice at all ("the craft is the bridge, the substrate is the rails"). DERIVED, bench-ready
3. **The sign-flip protocol as a community standard.** Most of this field's hundred-year noise floor exists because positives were reported without the §6 discriminator set. A drone company that publishes its balance design, its five controls, and its nulls — whatever the outcome — becomes the credible instrument this field has never had. That reputation is worth more than a marginal positive.
4. **The document hunt.** Three archival retrievals would move the §3 ledger more than any argument: the SNCASO "Project Montgolfier" report PDFs (released via Cornillon 2008 / Starburst 2012, primary site now offline — the target is the 1959 final report's force/current numbers, which no secondary source carries); the 1952 NRL "Appraisal of the T.T. Brown Electrogravity Device" (reported missing from the laboratory); and the 1968 Northrop electro-aerodynamics paper (drag

reduction, not thrust — but a real claimed AIAA item reported vanished). Each is a bounded task with a binary outcome. Relatedly: Brown's surviving *electrometer logs* (family/biographer archive) reportedly show anomalies patterned on the sidereal calendar — the §6 sidereal-folding discriminator adjudicates that claim mechanically if the logs surface, and either outcome is informative.

9. The Electrogravitics Claim

> 1. The force Townsend Brown measured in air is ion wind — conventional, useful, atmospheric, and not gravitational. MEASURED
> 2. Every published, independently instrumented vacuum test of the Brown effect is a null, and the tightest bounds the static-capacitor coupling at $\sim 2\times10^{-10}$. The one claimed vacuum positive — Project Montgolfier, 1955–57 — is documented but unquantified in the surviving record, and stays open until its numbers are retrieved or the §6 bench speaks. MEASURED (published record) / OPEN (the French residual)
> 3. The mechanism exists unconditionally: electromagnetic fields curve spacetime (Einstein–Maxwell, $8\pi G/c^4$; exact lab-hardware solutions, Füzfa 2016). The companion framework derives a specific force law — $F = C\cdot\kappa\cdot U\_drive/L\_grad$ — in which Brown's geometry is the right shape and coherence is the proposed missing ingredient; weak-field gravity itself is closed by the underlying law with receipts. THEOREM (existence) / DERIVED (form)
> 4. That law's coupling constant C is the single open number. It has never been measured above zero. Hover requires $\sim 10^{-6}$ of full coherence; the bare incoherent floor is $\sim 10^{-43}$. OPEN
> 5. Hover, if the coupling exists, is a near-zero-power static lock; propellantless is not reactionless; symmetric drives give zero; over-unity is forbidden. DERIVED
> 6. The decisive experiment is drone-shop-sized: one asymmetric device, one vacuum balance, five controls (§6). A positive that survives them changes everything; a clean null is the best bound ever published. OPEN
> 7. No engineering claim in this lineage has survived audit to date — and this paper's own program is the audit it must survive. MEASURED

References & receipts

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- Hathaway, Cleveland & Bao, *Physica C* 385, 488 (2003) (Podkletnov null). Harris, *Found. Phys. Lett.* 12, 201 (1999) (Li–Torr refutation). Graham et al., *Physica C* 468, 383 (2008).
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- Füzfa, *Phys. Rev. D* 93, 024014 (2016) (exact EM-sourced curvature). Tajmar, Kößling & Neunzig, *Sci. Rep.* 14, 19427 (2024) (broad coupling sweep). Tajmar et al., *Front. Phys.* 10, 892215 (2022) (supercurrent bound). Ummarino & Gallerati, *EPJ C* 77, 549 (2017); Gallerati, *Front. Phys.* 10, 941858 (2022) (superconductor gravito-Maxwell). Survey: `papers/GRAVITY_ENGINEERING_DEEPDIVE_2026-08-13.md`.
- Framework receipts: `crypto_oracle/shape_F_gravity_closure_probe.py` (factor-2 ray trace); `runtime_receipts/unruh_zpf_anisotropy_pricing.json` (ZPF pricing); `crypto_oracle/universe_sim/ampere_tension_sim.py` + `papers/AMPERE_LONGITUDINAL_SECTOR_NOTE_2026-07-20.md` (hairpin); `papers/G_VERTEX_DERIVATION.md` ($g \leq 1/\sqrt{3}$); `papers/WARP_DRIVE_MATH_AUDIT.md` (force law + conservation audit); `papers/BIEFELD_BROWN_SIM_PREREG_2026-07-19.md` (frozen pre-reg) → `crypto_oracle/universe_sim/biefeld_brown_sim.py` + `runtime_receipts/biefeld_brown_run.json` + `papers/RESULTS_BIEFELD_BROWN_2026-08-13.md` (run, all gates passed — §6); `hover/buhler_cross_check.py`, `hover/capacitor_cross_check.py`.
- Documentary audit behind the §3 Montgolfier row and the §8 document hunt: `papers/MICHELS_BROWN_DOCUMENTARY_AUDIT_2026-08-13.md` (§1/§1a).

Changelog

- **2026-08-13** — draft 0.1, written and revised in one day: initial field guide; mechanism register raised to theorem level (Einstein–Maxwell / Füzfa 2016 exact solutions; three-layer claim structure); Biefeld–Brown retrodiction sim run against its July pre-registration and scored (all four frozen predictions, all three gates); Montgolfier documentary audit folded into the §3 ledger and the §8 document hunt; Buhler claim-decline recorded; coverage analysis of the published null program (the one untested cell) added to §4; over-unity boundary stated precisely.
- **2026-08-14** — draft 0.2: editorial pass — the day's addenda folded into the body prose; §2 restructured to lead with the Einstein–Maxwell theorem; audit-file provenance moved to References and this changelog. No claim, number, or tag changed.

Earlier drafts of companion papers called the lattice coupling x_v; the current canonical object is the vertex-sharing strength $g \le 1/\sqrt{3}$ with substrate aggregate C_sub·$|xv|^2$ kept as the engineering coefficient. This paper uses "the coupling, C" throughout and lets the companions own the formalism.